

Xingrui Chen

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EDUCATION

Duke University

M.S. in Mechanical Engineering, Robotics Concentration

Aug 2025 - May 2027 (Expected)

GPA: 3.85/4.00

Hefei University of Technology

B.Eng. in Mechanical Engineering

Sep 2021 - Jun 2025

GPA: 3.44/4.00

Honors: First Prize, China Formula Student Competition; First Prize, 3D Modeling Competition.

TECHNICAL SKILLS

CAD & Analysis: CATIA, SolidWorks, ANSYS (FEA), STAR-CCM+ (CFD), MATLAB.

Fabrication: Carbon-fiber wet layup, composite mold design, vacuum bagging, 3D printing.

Robotics & Testing: Hardware integration, workspace calibration, Isaac Lab simulation.

ENGINEERING EXPERIENCE

Formula Student | Aerodynamics & Bodywork Team Lead

Sep 2022 - Oct 2024

- Led five teammates across two seasons; designed front and rear wings and optimized the diffuser in CATIA.
- Improved simulated downforce-to-drag ratio by **16% (2.5 to 2.9)** using STAR-CCM+; CFD predicted **926 N of downforce at 16.67 m/s**.
- Designed machined PMI foam tooling; fabricated carbon-fiber wing components using wet layup and vacuum bagging.
- Designed hollow wings with aluminum trusses; developed skin layup and assembly using two-piece female molds.

Robot-Assisted Laser Lithotripsy | Experimental Systems & Validation

Jan 2026 - Present

- Integrated a DOBOT Nova 5 with a Ho:YAG laser testbed; programmed two- and four-step fiber-positioning protocols for underwater stone ablation at 0.4 J and 20 Hz.
- Developed MATLAB segmentation of 512^3 -voxel OCT scans to quantify crater volume, depth, and profile area.
- Four-step control yielded **44.5% greater crater volume** at 320 pulses versus fixed-standoff control (initial distance: 0.75 mm; $p = 0.00173$).

SO-101 Robotic Drawing | Calibration, Control & Tool-Wear Modeling

Apr 2026 - Present

- Built contour-fill and Bézier stroke planning for an SO-101 arm with an oil-pastel holder and overhead vision.
- Established **49-point calibration over a 120×120 mm workspace** using bilinear joint-angle interpolation; implemented visual checkpoints with image registration and bounded corrections to future strokes.
- Tuned joint PID gains, reducing aggregate joint-angle **MAE by 47% (0.527° to 0.279°)** and **P90 error by 67%** across 120 positioning measurements.
- Derived a mass-loss model for fixed-angle pastel wear and tool-height recession; experimental validation is ongoing.

CEER | Humanoid Simulation, Prototyping & Testing

Jan 2026 - Apr 2026

- Built room-scale **Isaac Sim** scenes with randomized furniture layouts and box positions; tested humanoid box grasping, transport, and placement for CEER system evaluation.
- Contributed to end-effector design and 3D printing; completed **10+ hours of hardware testing and data collection**.

Humanoid Waist-Joint RV Reducer | Mechanical Design & FEA

Oct 2024 - May 2025

- Designed a two-stage RV reducer in SolidWorks for a **$220 \text{ N} \cdot \text{m}$ output target** and **180 mm diameter limit**; sized the transmission and selected a coaxial frameless motor.
- Redesigned the pin housing for interference-fit integration with the joint enclosure; produced **assembly and six part drawings** and planned machining and heat-treatment routes.
- Evaluated first-stage spur gears with ANSYS transient contact FEA under **$3 \text{ N} \cdot \text{m}$ per driven gear**, using **0.5 mm mesh refinement** at tooth contacts to assess stress and deformation.

SELECTED PUBLICATIONS

X. Luo, X. Chen, et al. "CEER: Compliant End-Effector and Root Control as a Unified Interface for Hierarchical Humanoid Loco-Manipulation." *IEEE/RSJ IROS*, 2026.

X. Chen. "Analysis and Improvements of a Front Wing for Formula SAE Car." *CIAP*, 2024.